

Extremal Normality of Trees with a Prescribed Number of Leaves

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Abstract

Let G be a connected graph. The normality of a vertex v is

$$\text{norm}_G(v) = \min\{d_G(v, u) : u \in \mathcal{P}(G)\},$$

where $\mathcal{P}(G)$ is the set of peripheral vertices of G , and $\text{Norm}(G) = \sum_{v \in V(G)} \text{norm}_G(v)$ is the total normality. We determine both the minimum and the maximum of Norm among all trees of order n with exactly k leaves. The minimum has three cases according to the remainder of $n - 1$ modulo k , and all equality trees are described by level counts from the centre. For the maximum, the boundary cases $k = 2$, $k = n - 2$ and $k = n - 1$ are explicit. In the remaining range $3 \leq k \leq n - 3$, every maximizer has exactly two peripheral vertices. The maximum is then obtained by taking the largest value, over an explicit finite set of feasible diameters, of a closed formula involving a rooted profile function. The extremal trees are precisely a diametral path with all remaining leaves placed at the centre according to a deepest-level rooted profile.

Keywords. normality; peripheral vertex; leaf; tree; extremal tree; diameter.

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1 Introduction

All graphs considered in this paper are finite, simple and connected. For a graph $G = (V(G), E(G))$, the distance between vertices u and v is denoted by $d_G(u, v)$. The eccentricity of v is

$$\text{ecc}_G(v) = \max_{u \in V(G)} d_G(v, u).$$

The diameter and radius of G are

$$\text{diam}(G) = \max_{v \in V(G)} \text{ecc}_G(v), \quad \text{rad}(G) = \min_{v \in V(G)} \text{ecc}_G(v).$$

A vertex v is called *peripheral* if $\text{ecc}_G(v) = \text{diam}(G)$; the set of peripheral vertices is denoted by $\mathcal{P}(G)$. The normality of v is the distance from v to the nearest peripheral vertex,

$$\text{norm}_G(v) = \min_{u \in \mathcal{P}(G)} d_G(v, u),$$

and the total normality of G is

$$\text{Norm}(G) = \sum_{v \in V(G)} \text{norm}_G(v).$$

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Thus peripheral vertices contribute zero, while vertices far from all peripheral vertices contribute more.

The parameter normality was introduced by Chen, Wang and Zhang [1]. They studied extremal problems for trees under several constraints, including fixed diameter and fixed number of peripheral vertices. Li, Liu and Zhang [3] later studied related extremal problems with prescribed matching number. In a tree every peripheral vertex is a leaf, but a leaf need not be peripheral. Therefore prescribing the number of leaves is a natural condition different from prescribing the number of peripheral vertices.

Let $\ell(T)$ be the number of leaves of a tree T . We write

$$\mathcal{T}_{n,k} = \{T : T \text{ is a tree of order } n \text{ and } \ell(T) = k\}.$$

This paper determines

$$\min\{\text{Norm}(T) : T \in \mathcal{T}_{n,k}\} \quad \text{and} \quad \max\{\text{Norm}(T) : T \in \mathcal{T}_{n,k}\}$$

for all $2 \leq k \leq n-1$, and characterizes all extremal trees. The minimum problem is controlled by nearest-leaf layers and by a sharp theorem of Qiao and Zhan [4] on the minimum possible diameter of a tree with prescribed order and number of leaves. The maximum problem has a different structure: apart from the star, the double-star and the path cases, every maximizer has exactly two peripheral leaves. Hence the other $k-2$ leaves must be kept non-peripheral and placed as far from the two peripheral leaves as possible. This reduces the problem to a rooted profile maximization at the centre of a diametral path.

2 Preliminaries

The path and star on n vertices are denoted by P_n and S_n , respectively. For a tree T , a leaf is a vertex of degree one, and the set of leaves is denoted by $L(T)$. If $X \subseteq V(T)$ and $v \in V(T)$, write $d_T(v, X) = \min\{d_T(v, x) : x \in X\}$.

Proposition 2.1. *Every peripheral vertex of a nontrivial tree is a leaf.*

Proof. Let T be a tree with diameter D , and let $x \in \mathcal{P}(T)$. Choose a vertex y such that $d_T(x, y) = D$. If x is not a leaf, then x has a neighbour z which is not the next vertex on the unique x - y path. The unique z - y path first goes from z to x and then follows the x - y path. Therefore

$$d_T(z, y) = 1 + d_T(x, y) = D + 1,$$

which contradicts the definition of D . Hence x is a leaf. □

The converse of Proposition 2.1 is false: a leaf attached near the middle of a long path need not be peripheral.

We shall use the following theorem of Qiao and Zhan.

Lemma 2.2 (Qiao–Zhan [4]). *Let $T \in \mathcal{T}_{n,k}$ with $k \geq 2$. Write*

$$n - 1 = zk + r, \quad z \geq 1, \quad 0 \leq r < k.$$

Then

$$\text{diam}(T) \geq D_{\min}(n, k) = \begin{cases} 2z, & r = 0, \\ 2z + 1, & r = 1, \\ 2z + 2, & 2 \leq r < k. \end{cases}$$

For the minimum problem it is useful to compare normality with the nearest-leaf distance

$$u_T(v) = d_T(v, L(T)), \quad U(T) = \sum_{v \in V(T)} u_T(v).$$

Since $\mathcal{P}(T) \subseteq L(T)$ by Proposition 2.1, we have

$$\text{norm}_T(v) \geq u_T(v) \quad \text{for all } v \in V(T), \quad \text{and hence} \quad \text{Norm}(T) \geq U(T). \quad (1)$$

Moreover, equality in (1) for every vertex holds if and only if every leaf of T is peripheral.

Lemma 2.3 (Leaf-distance layers). *Let T be a tree with k leaves. For $j \geq 0$, put*

$$W_j(T) = \{v \in V(T) : u_T(v) = j\}.$$

Then $|W_0(T)| = k$ and $|W_j(T)| \leq k$ for every $j \geq 1$.

Proof. The set $W_0(T)$ is exactly $L(T)$, so $|W_0(T)| = k$. Fix $j \geq 1$ and put $H = T[\{v : u_T(v) \leq j\}]$. Every vertex in $W_j(T)$ has at least one neighbour outside H , namely the next vertex on a shortest path from it to a leaf. Delete all edges joining $W_j(T)$ to $V(T) \setminus V(H)$. Each deleted edge leads to a component outside H that contains at least one original leaf of T , and different deleted edges lead to disjoint such components; otherwise a cycle would be formed in T . Hence the number of deleted edges is at most k . Since each vertex of $W_j(T)$ is incident with at least one deleted edge, $|W_j(T)| \leq k$. $\square$

Lemma 2.4 (Minimum of the nearest-leaf distance). *Let $T \in \mathcal{T}_{n,k}$, where $2 \leq k \leq n-1$, and write $n-1 = zk + r$ with $z \geq 1$ and $0 \leq r < k$. Then*

$$U(T) \geq B(n, k) := k \frac{z(z-1)}{2} + (r+1)z. \quad (2)$$

Equality holds in (2) if and only if T is obtained from a core tree C of order $r+1$ by attaching k pendent paths of length z to vertices of C , with every vertex of C receiving at least one such path.

Proof. By Lemma 2.3, there are exactly k vertices in $W_0(T)$ and at most k vertices in each later layer. To minimize the sum of the layer indices over all n vertices, one first fills W_0 , then fills W_1 , and so on. Since

$$n = k + k(z-1) + (r+1),$$

at least $r+1$ vertices have nearest-leaf distance at least z . Therefore

$$U(T) \geq k(1 + 2 + \cdots + (z-1)) + (r+1)z = k \frac{z(z-1)}{2} + (r+1)z.$$

Equality is possible only when

$$|W_0| = |W_1| = \cdots = |W_{z-1}| = k, \quad |W_z| = r+1,$$

and no vertex lies in a layer W_j with $j > z$. The vertices in W_z induce a tree C of order $r+1$. Every vertex of C must be incident with at least one path of length z ending in a leaf; otherwise it would not be at distance z from the nearest leaf. Since the first z layers are full, these attached paths account for all k leaves and are pairwise internally disjoint. This gives the stated structure. Conversely, every tree with that structure has the displayed layer sizes and attains equality. $\square$

3 Minimum normality

Write

$$n - 1 = zk + r, \quad z \geq 1, \quad 0 \leq r < k. \quad (3)$$

Definition 3.1 (The family $\mathcal{F}_{n,k}$). *Let $2 \leq k \leq n - 1$, and let z, r be defined by (3). The family $\mathcal{F}_{n,k}$ is defined as follows.*

- (a) *If $r = 0$, then $\mathcal{F}_{n,k}$ consists of the spider with one centre and k legs, each of length z .*
- (b) *If $r = 1$, then $\mathcal{F}_{n,k}$ consists of the trees obtained from a central edge c_1c_2 by attaching p internally disjoint paths of length z to c_1 and $k - p$ internally disjoint paths of length z to c_2 , where $1 \leq p \leq k - 1$.*
- (c) *If $2 \leq r < k$, then $\mathcal{F}_{n,k}$ consists of the rooted trees with centre c whose first level has exactly r vertices, whose levels $2, 3, \dots, z + 1$ each have exactly k vertices, and whose leaves are precisely the vertices at level $z + 1$. Equivalently, the r first-level vertices split the k terminal paths into r nonempty groups.*

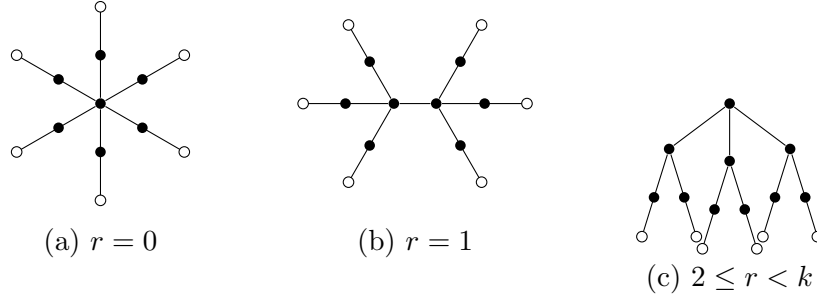


Figure 1: The three minimum-normality extremal types. Open vertices denote leaves.

Theorem 3.2. *Let $T \in \mathcal{T}_{n,k}$, where $2 \leq k \leq n - 1$, and write $n - 1 = zk + r$ with $z \geq 1$ and $0 \leq r < k$. Then*

$$\text{Norm}(T) \geq M(n, k),$$

where

$$M(n, k) = \begin{cases} k \frac{z(z-1)}{2} + z, & r = 0, \\ k \frac{z(z-1)}{2} + 2z, & r = 1, \\ k \frac{z(z-1)}{2} + (r+1)z + 1, & 2 \leq r < k. \end{cases} \quad (4)$$

Equality holds if and only if $T \in \mathcal{F}_{n,k}$.

Proof. Let $B(n, k)$ be the lower bound in Lemma 2.4. From (1) and Lemma 2.4,

$$\text{Norm}(T) \geq U(T) \geq B(n, k) = k \frac{z(z-1)}{2} + (r+1)z. \quad (5)$$

If $r = 0$, then $B(n, k) = kz(z-1)/2 + z$. The tree described in Definition 3.1(a) has all leaves peripheral, so $\text{Norm} = U$, and it attains this value. If equality holds in (5), then $\text{Norm}(T) = U(T)$ and equality holds in Lemma 2.4. Hence every leaf is peripheral and the core in Lemma 2.4 has one vertex. Thus T is exactly the spider in Definition 3.1(a).

If $r = 1$, then $B(n, k) = kz(z-1)/2 + 2z$. The trees in Definition 3.1(b) have all leaves peripheral and attain this value. Conversely, equality in (5) again implies $\text{Norm}(T) = U(T)$

and equality in Lemma 2.4. The core has two vertices; since every core vertex receives at least one path of length z , the tree is exactly one of the central-edge trees in Definition 3.1(b).

It remains to consider $2 \leq r < k$. Here $B(n, k) = kz(z-1)/2 + (r+1)z$. We first show that the lower bound must be improved by one. Suppose that $U(T) = B(n, k)$. By Lemma 2.4, T is obtained from a core tree C of order $r+1 \geq 3$ by attaching k paths of length z , with every core vertex receiving at least one path. A leaf attached to a core vertex c is peripheral in T if and only if c is peripheral in C . Since a tree of order at least three has a non-peripheral vertex, choose $c \in V(C) \setminus \mathcal{P}(C)$. At least one length- z pendent path is attached at c . For each of the z vertices on this path and for c itself, the difference $\text{norm}_T - u_T$ is at least $d_C(c, \mathcal{P}(C)) \geq 1$. Consequently

$$\text{Norm}(T) - U(T) \geq z + 1 \geq 2.$$

Thus a tree with $U(T) = B(n, k)$ has $\text{Norm}(T) \geq B(n, k) + 2$. If instead $U(T) \geq B(n, k) + 1$, then $\text{Norm}(T) \geq B(n, k) + 1$. Therefore

$$\text{Norm}(T) \geq B(n, k) + 1 = k \frac{z(z-1)}{2} + (r+1)z + 1.$$

If $\text{Norm}(T) = B(n, k) + 1$, then the previous paragraph shows that $U(T) \neq B(n, k)$; hence $U(T) = B(n, k) + 1$ and $\text{Norm}(T) = U(T)$. Therefore every leaf of T is peripheral. By Lemma 2.2, $\text{diam}(T) \geq 2z + 2$. Since all leaves are peripheral, a middle vertex of a diametral path has nearest-leaf distance at least $z + 1$. The equality $U(T) = B(n, k) + 1$ then forces the leaf-distance layers to be

$$|W_0| = |W_1| = \dots = |W_{z-1}| = k, \quad |W_z| = r, \quad |W_{z+1}| = 1,$$

and no other layers occur. Let c be the unique vertex in W_{z+1} . Then the vertices in W_z are exactly the r neighbours of c , and the levels $2, 3, \dots, z+1$ measured from c each contain exactly k vertices. Thus T is one of the trees in Definition 3.1(c). Conversely, every tree in Definition 3.1(c) has all leaves peripheral and has normality

$$(z+1) + rz + k(1 + 2 + \dots + (z-1)) = k \frac{z(z-1)}{2} + (r+1)z + 1.$$

This proves the theorem. □

4 A rooted profile lemma

The maximum problem will be reduced to a rooted depth-sum problem. Let R be a rooted tree with root ρ . The root is not counted as a leaf in the following definition. Suppose that R has L leaves, $L + t$ non-root vertices and height at most m . For a non-root vertex x , let $\delta_R(x) = d_R(\rho, x)$, and define

$$\Phi(L, t, m) = \max_R \sum_{x \in V(R) \setminus \{\rho\}} \delta_R(x), \quad (6)$$

where the maximum is taken over all such rooted trees.

Lemma 4.1 (Rooted profile lemma). *Let $L \geq 1$, $m \geq 1$, and $0 \leq t \leq L(m-1)$. Then $\Phi(L, t, m)$ is as follows.*

- (i) *If $0 \leq t \leq m-1$, then an extremal rooted tree is a comet: it has a common trunk of t internal vertices and all L leaves are attached to the last vertex of the trunk. In this case*

$$\Phi(L, t, m) = \frac{t(t+1)}{2} + L(t+1). \quad (7)$$

(ii) If $m - 1 \leq t \leq L(m - 1)$ and $L \geq 2$, write

$$E = t - m + 1 = q(L - 1) + s, \quad 0 \leq s < L - 1. \quad (8)$$

Then

$$\Phi(L, t, m) = mL + \frac{m(m - 1)}{2} + (L - 1) \left(qm - \frac{q(q + 1)}{2} \right) + s(m - q - 1). \quad (9)$$

For $L = 1$, only case (i) can occur.

Proof. For a rooted tree with L leaves and height at most m , every level has at most L vertices: each vertex at a given level must have at least one descendant leaf, and the descendant-leaf sets of distinct vertices at the same level are disjoint.

If $0 \leq t \leq m - 1$, all non-root internal vertices can lie on one common trunk. Branching before all t internal vertices have been used only makes some leaves shallower; moving the branching point away from the root increases the depth of each leaf below it and does not decrease any other depth. Hence the comet is extremal, giving

$$(1 + 2 + \cdots + t) + L(t + 1) = \frac{t(t + 1)}{2} + L(t + 1).$$

Now assume $t \geq m - 1$ and $L \geq 2$. First put one vertex in each level $1, 2, \dots, m - 1$ and all L leaves in level m . This uses $m - 1$ internal vertices. The number of remaining internal vertices is $E = t - (m - 1) = t - m + 1$. To maximize the depth sum, fill the deepest available levels first. Each level $j < m$ has additional capacity $L - 1$. Thus, if $E = q(L - 1) + s$ with $0 \leq s < L - 1$, then q levels are filled completely, starting from level $m - 1$, and then s vertices are placed in the next level. This yields (9). $\square$

Remark 4.2. The equality profile in Lemma 4.1 may be described by level sizes. In case (i), the nonzero level sizes are

$$b_1 = b_2 = \cdots = b_t = 1, \quad b_{t+1} = L.$$

In case (ii), start with $b_1 = \cdots = b_{m-1} = 1$ and $b_m = L$. Then fill the levels $m - 1, m - 2, \dots$ up to size L , using the remaining $E = t - m + 1$ internal vertices. Any rooted tree with this level profile and with every vertex in a level below the last having at least one child in the next level is extremal.

5 Maximum normality

We now solve the maximum problem on $\mathcal{T}_{n,k}$. The path case and the two very leafy boundary cases are immediate.

Lemma 5.1. The following assertions hold.

(i) If $k = 2$, then $\mathcal{T}_{n,2} = \{P_n\}$ and

$$\max_{T \in \mathcal{T}_{n,2}} \text{Norm}(T) = \left\lfloor \frac{(n - 1)^2}{4} \right\rfloor.$$

(ii) If $k = n - 1$, then $\mathcal{T}_{n,n-1} = \{S_n\}$ and

$$\max_{T \in \mathcal{T}_{n,n-1}} \text{Norm}(T) = 1.$$

(iii) If $k = n - 2 \geq 3$, then every tree in $\mathcal{T}_{n,k}$ is a double-star, and

$$\max_{T \in \mathcal{T}_{n,n-2}} \text{Norm}(T) = 2.$$

All double-stars with $n - 2$ leaves attain equality.

Proof. A tree with two leaves is a path, and the two leaves of P_n are the only peripheral vertices. Hence

$$\text{Norm}(P_n) = \sum_{i=0}^{n-1} \min\{i, n-1-i\} = \left\lfloor \frac{(n-1)^2}{4} \right\rfloor.$$

A tree with $n - 1$ leaves is the star S_n , whose centre has normality one and whose leaves are all peripheral. If $k = n - 2$, then the tree has exactly two non-leaves. The two non-leaves are adjacent, all other vertices are leaves adjacent to one of them, and the tree is a double-star. Its leaves are all peripheral and its two non-leaves have normality one. Thus $\text{Norm} = 2$. $\square$

The next lemma is the key structural step. It says that, away from the boundary cases in Lemma 5.1, a maximum-normality tree has as few peripheral vertices as possible.

Lemma 5.2 (Two peripheral leaves in a maximizer). *Let $3 \leq k \leq n - 3$, and let $T \in \mathcal{T}_{n,k}$ maximize Norm . Then $|\mathcal{P}(T)| = 2$.*

Proof. By Proposition 2.1, $|\mathcal{P}(T)| \geq 2$. Suppose, to the contrary, that $|\mathcal{P}(T)| \geq 3$. Let $D = \text{diam}(T)$. Since T is neither a star nor a double-star, $D \geq 4$.

We first choose three peripheral leaves x, y, z such that

$$d_T(x, y) = d_T(z, y) = D. \quad (10)$$

This is always possible from the usual centre description of a tree. If D is even, peripheral leaves are exactly the leaves at maximum distance from the unique centre. Choose a maximum branch whose complement contains at least two peripheral leaves, take y in that branch, and take x, z outside it. If D is odd, remove the central edge. Peripheral leaves occur at maximum depth on the two sides; choose y on one side and choose x, z on the other side, taking the side with at least two peripheral leaves as the side of x, z .

Let z^- be the neighbour of z . We distinguish two cases.

First suppose that $\deg_T(z^-) \geq 3$. Delete the edge z^-z and join z to a central vertex of the diametral path from x to y ; when the centre is an edge, choose either of its two ends. Call the resulting tree T_1 . The number of leaves is unchanged: z remains a leaf, z^- is still not a leaf, and the chosen central vertex is not a leaf. The vertices x and y are still at distance D . Also, the moved leaf z is now at distance at most $1 + \lceil D/2 \rceil < D$ from every vertex, because $D \geq 4$. Thus no new peripheral vertex is created and z is no longer peripheral. Hence $\mathcal{P}(T_1) \subseteq \mathcal{P}(T) \setminus \{z\}$. Distances from all old vertices other than z to the remaining peripheral vertices are unchanged, so their normalities do not decrease. The vertex z changes from a peripheral vertex to a non-peripheral leaf and has positive normality in T_1 . Therefore $\text{Norm}(T_1) > \text{Norm}(T)$, a contradiction.

It remains to consider the case $\deg_T(z^-) = 2$. Delete the edge z^-z and add the edge xz ; call the resulting tree T_2 . The leaf x becomes an internal vertex, the vertex z^- becomes a leaf, and z remains a leaf, so the number of leaves is again unchanged. By (10), the new distance from z to y is $D + 1$, and no distance can exceed $D + 1$. Thus z and y are peripheral in T_2 . For every old vertex $v \neq z$, the distance from v to the new peripheral vertex z is $d_T(v, x) + 1$, which is larger than its old distance to the peripheral vertex x . The old peripheral vertex z has been removed from its former position, and all other old peripheral vertices either remain at the same distances or cease to be peripheral. Consequently $\text{norm}_{T_2}(v) \geq \text{norm}_T(v)$ for every $v \neq z$, while $\text{norm}_{T_2}(x) = 1 > 0 = \text{norm}_T(x)$. The moved vertex z has normality zero both before and after the operation. Hence $\text{Norm}(T_2) > \text{Norm}(T)$, again a contradiction.

Therefore a maximizer cannot have more than two peripheral vertices. $\square$

For a tree whose only peripheral vertices are the endpoints of a diametral path, the normality is controlled by the centre of that path. Fix a diametral path

$$P = v_0 v_1 \cdots v_d$$

and set

$$h = \left\lfloor \frac{d}{2} \right\rfloor, \quad L = k - 2, \quad t = n - d - k + 1. \quad (11)$$

Here L is the number of leaves outside P , and $L + t = n - (d + 1)$ is the number of vertices outside P . The path itself contributes

$$F(d) = \sum_{i=0}^d \min\{i, d - i\} = \left\lfloor \frac{d^2}{4} \right\rfloor. \quad (12)$$

Indeed, if $d = 2h$, then $F(d) = h^2$, while if $d = 2h + 1$, then $F(d) = h(h + 1)$.

Lemma 5.3 (Fixed diameter with two peripheral vertices). *Let $3 \leq k \leq n - 3$, and let $T \in \mathcal{T}_{n,k}$ have diameter d and exactly two peripheral vertices. With h, L, t as in (11), one has*

$$0 \leq t \leq L(h - 2), \quad (13)$$

and

$$\text{Norm}(T) \leq \left\lfloor \frac{d^2}{4} \right\rfloor + h(L + t) + \Phi(L, t, h - 1). \quad (14)$$

Equality holds precisely for the trees obtained by attaching an extremal rooted profile from Lemma 4.1, with $m = h - 1$, at the centre of a diametral path. If d is even, the root is identified with the unique central vertex v_h . If d is odd, the root may be split across the central edge $v_h v_{h+1}$: each child of the formal root is attached to one of v_h and v_{h+1} .

Proof. Let the two peripheral vertices be v_0 and v_d on the diametral path $P = v_0 v_1 \cdots v_d$. The vertices of P contribute $F(d)$ in (12).

Consider a component outside P attached at v_i , and let x be a vertex of this component at distance δ from v_i . Since the only peripheral vertices are v_0 and v_d ,

$$\text{norm}_T(x) = \delta + \min\{i, d - i\} \leq \delta + h. \quad (15)$$

If x is a leaf outside P , then it is not peripheral. Hence

$$\delta + \max\{i, d - i\} < d,$$

so $\delta \leq \min\{i, d - i\} - 1 \leq h - 1$. Thus all outside leaves have depth at most $h - 1$ from their attachment points. The outside part has L leaves and $L + t$ vertices; since a rooted tree of height at most $h - 1$ with L leaves has at most $L(h - 1)$ non-root vertices, we obtain $t \leq L(h - 2)$. The inequality $t \geq 0$ is just the fact that the L leaves outside P already require L outside vertices.

Now identify the attachment points of all outside components with a formal root. This gives a rooted tree with L leaves, $L + t$ non-root vertices and height at most $h - 1$; the sum of the depths of the non-root vertices is unchanged. By (15), the outside contribution to $\text{Norm}(T)$ is at most

$$h(L + t) + \Phi(L, t, h - 1).$$

Adding the path contribution gives (14).

Equality in (15) requires every outside component to be attached at a central vertex; for odd d , either central vertex gives the same value h . Equality in the depth-sum part requires exactly an extremal rooted profile from Lemma 4.1. The described trees have only v_0 and v_d as peripheral vertices, because every outside leaf has depth at most $h - 1$ from the central set, and they attain the displayed bound. $\square$

We can now state the maximum theorem. For $3 \leq k \leq n - 3$, define the set of feasible diameters

$$\mathcal{D}_{n,k} = \left\{ d \in \mathbb{Z} : 4 \leq d \leq n - k + 1, 0 \leq n - d - k + 1 \leq (k - 2) \left(\left\lfloor \frac{d}{2} \right\rfloor - 2 \right) \right\}. \quad (16)$$

The set $\mathcal{D}_{n,k}$ is nonempty, since $d = n - k + 1 \geq 4$ belongs to it. For $d \in \mathcal{D}_{n,k}$, put

$$\Gamma(n, k, d) = \left\lfloor \frac{d^2}{4} \right\rfloor + \left\lfloor \frac{d}{2} \right\rfloor (n - d - 1) + \Phi \left(k - 2, n - d - k + 1, \left\lfloor \frac{d}{2} \right\rfloor - 1 \right). \quad (17)$$

The term $n - d - 1$ is the number of vertices outside the diametral path, equivalently $L + t$.

Let

$$\mathcal{A}_{n,k} = \{d \in \mathcal{D}_{n,k} : \Gamma(n, k, d) = \max_{e \in \mathcal{D}_{n,k}} \Gamma(n, k, e)\}. \quad (18)$$

For $d \in \mathcal{A}_{n,k}$, let $\mathcal{H}_{n,k}(d)$ be the family of trees constructed as follows. Start with a path $v_0 v_1 \cdots v_d$. Put $h = \lfloor d/2 \rfloor$, $L = k - 2$, $t = n - d - k + 1$ and $m = h - 1$. Attach to the centre of this path an extremal rooted profile attaining $\Phi(L, t, m)$. If d is even, the root is identified with v_h . If d is odd, the root is replaced by the central edge $v_h v_{h+1}$ and the children of the formal root are distributed arbitrarily between v_h and v_{h+1} .

Theorem 5.4 (Maximum normality). *Let $2 \leq k \leq n - 1$.*

(i) *If $k = 2$, then*

$$\max_{T \in \mathcal{T}_{n,2}} \text{Norm}(T) = \left\lfloor \frac{(n-1)^2}{4} \right\rfloor,$$

and equality holds only for P_n .

(ii) *If $k = n - 1$, then*

$$\max_{T \in \mathcal{T}_{n,n-1}} \text{Norm}(T) = 1,$$

and equality holds only for S_n .

(iii) *If $k = n - 2 \geq 3$, then*

$$\max_{T \in \mathcal{T}_{n,n-2}} \text{Norm}(T) = 2,$$

and equality holds for all double-stars with $n - 2$ leaves.

(iv) *If $3 \leq k \leq n - 3$, then*

$$\max_{T \in \mathcal{T}_{n,k}} \text{Norm}(T) = \max_{d \in \mathcal{D}_{n,k}} \Gamma(n, k, d). \quad (19)$$

Moreover, equality holds exactly for the trees in

$$\bigcup_{d \in \mathcal{A}_{n,k}} \mathcal{H}_{n,k}(d).$$

Proof. Parts (i)–(iii) are Lemma 5.1. Now assume $3 \leq k \leq n - 3$. Let $T \in \mathcal{T}_{n,k}$ be a maximizer. By Lemma 5.2, T has exactly two peripheral vertices. Let $d = \text{diam}(T)$. Lemma 5.3 implies $d \in \mathcal{D}_{n,k}$ and

$$\text{Norm}(T) \leq \Gamma(n, k, d) \leq \max_{e \in \mathcal{D}_{n,k}} \Gamma(n, k, e).$$

Conversely, for each $d \in \mathcal{D}_{n,k}$, the construction in $\mathcal{H}_{n,k}(d)$ gives a tree in $\mathcal{T}_{n,k}$ with exactly two peripheral vertices and normality $\Gamma(n, k, d)$. Hence the upper bound is sharp, and (19) follows. The equality statement is exactly the equality statement in Lemma 5.3, together with the requirement that the chosen diameter d belong to the argmax set $\mathcal{A}_{n,k}$. $\square$

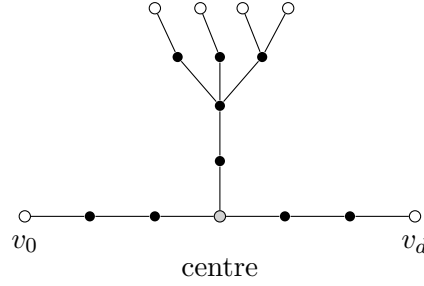


Figure 2: A typical maximum-normality tree in the main range $3 \leq k \leq n - 3$.

Example 5.5. Let $n = 20$ and $k = 5$. Then $L = k - 2 = 3$. The feasible diameters are $d = 10, 11, 12, 13, 14, 15, 16$. Evaluating (17) gives

d	10	11	12	13	14	15	16
$\Gamma(20, 5, d)$	96	94	103	96	96	91	91.

Thus the maximum is 103, attained only for $d = 12$. Here $h = 6$, $t = 20 - 12 - 5 + 1 = 4$ and $m = 5$. Since $t \leq m - 1$, Lemma 4.1(i) applies: the extremal tree is obtained from a path of length 12 by attaching at its centre a comet with a trunk of four internal vertices and three leaves at the end of the trunk.

6 Small-order verification

The proofs above are theoretical. Still, small-order enumerations are useful for detecting boundary mistakes. The following table records representative checks of the maximum theorem by enumerating non-isomorphic trees. The column d^* gives the maximizing diameter or diameters.

n	k	maximum Norm	d^*	extremal type from Theorem 5.4
6	3	7	4	path plus one central leaf
7	3	9	5	odd diametral path plus one central leaf
8	4	12	5	odd diametral path plus two central leaves
9	3	18	6	central comet with one extra internal vertex
10	4	23	6	central comet with two outside leaves
11	5	28	6	central comet with three outside leaves
12	6	33	6	central comet with four outside leaves

For the minimum theorem, representative checks include the following. The column “types” is the number of non-isomorphic equality types.

n	k	z	r	$M(n, k)$	equality types
7	3	2	0	5	1
8	3	2	1	7	1
9	3	2	2	10	1
10	4	2	1	8	2
11	4	2	2	11	2
12	5	2	1	9	2

7 Conclusion

We determined the exact minimum and maximum total normality among all trees of order n with k leaves. The minimum is governed by nearest-leaf layers and by the minimum possible diameter of a tree with given order and number of leaves. The maximum has a different extremal mechanism: except for the path, star and double-star boundary cases, a maximizer has exactly two peripheral vertices, and all other leaves are arranged near the centre of a diametral path according to a deepest-level rooted profile. The final maximum value is the explicit finite maximum in (19), and all equality trees are the corresponding central-profile constructions.

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